Polymers

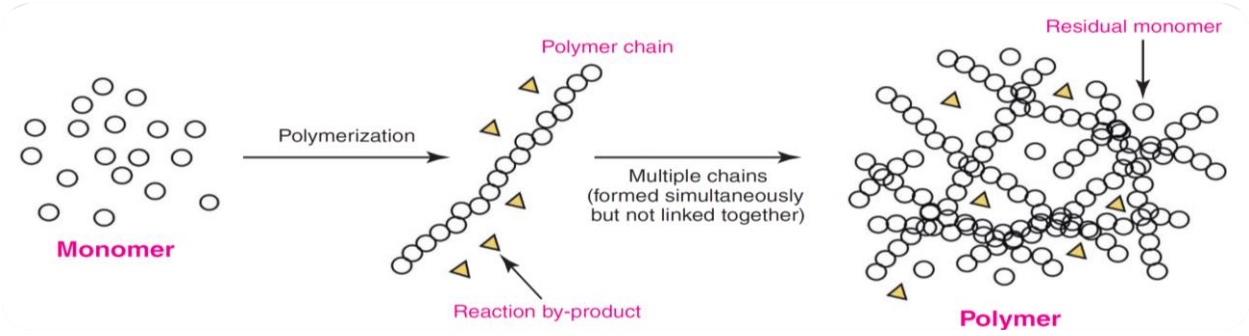
Polymer: a chemical compound consisting of a large organic molecule (macromolecule) formed by the union of many smaller repeating units (**mers**).

Monomer: it is a single molecule from which the polymer is constructed. It is a chemical compound that is capable of reacting to form a polymer.

Mer: The term used to designate the repeating unit or units in a polymer chain; thus mers are the "links" in the chain.

Polymerization: Chemical reaction in which monomers of a low molecular weight are converted into chains of polymers with a high molecular weight. The polymerization is *exothermic reaction*.

Monomer+ Monomer+ Monomer+ Monomer —polymerization→ Mer —— Mer —— Mer —— Mer ——



Copolymers: Polymer made of two or more monomer species and identical monomer units (mers) occurring in relatively long sequences along the main polymer chain. Copolymerization is used to improve the physical properties of the resulting resin.

Crosslinked: A difunctional or multifunctional monomer that forms a link between two polymer chains. Crosslinked polymers have many such crosslinks between neighboring chains such that a three-dimensional interconnected polymer network result.

Degree of polymerization (D.P): is specified by the total number of repeated units in the chain of the polymer. The higher the degree of polymerization the better will be the physical properties.

Glass transition temperature (Tg): The temperature at which macromolecule molecular motion begins to force the polymer chains apart. Thus, polymeric materials soften when heated above this temperature (It is the temperature was the polymers leave the rigid state and transfer into rubber like phase).

Factor control polymer properties:

1. ***Chemical composition of the polymer*:** This depends on types of monomers and its structure. e.g.

CH_2—CH---R vinyl polymer

when R=H polyethylene, the repeating unit is CH_2 group. Polyethylene is hydrophobic semicrystalline polymer.

But when R=OH the (polyvinyl alcohol) hydrophilic, water soluble polymer with gelling properties.

2. ***Topology of polymer chain:***

- **Linear polymer**. A-A-A-A

- **Nonlinear (branched) polymer**.

```
                A-A-A
A-A-A-A-A-A-
  \
   A-A-A
```

- **Cross linked polymer (polymer network).** The cross linking (permanent connection between the polymer chains) restricting the motion of the chains and improve rigidity of polymer.

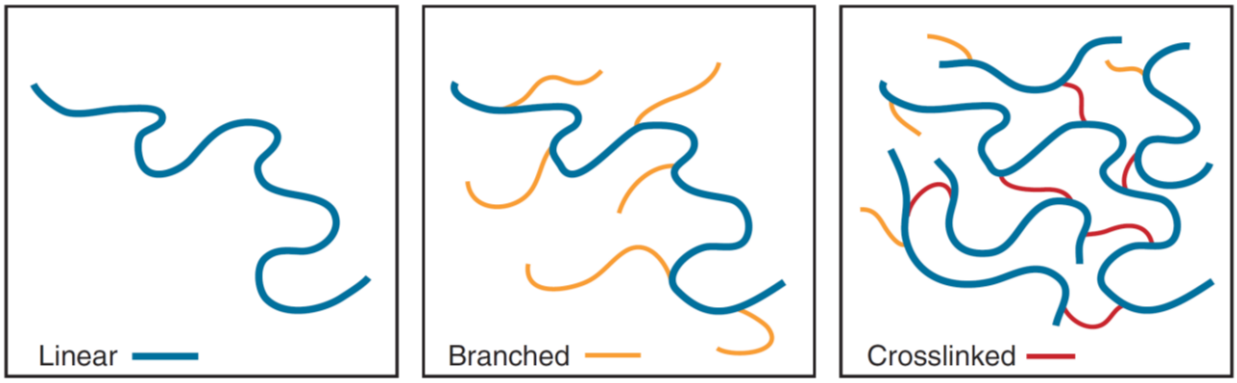
A-A-A-A-A-A-A

A-A-A-A-A-A-A-A

A-A-A-A-A-A-A-

The advantages of cross-linking agents:

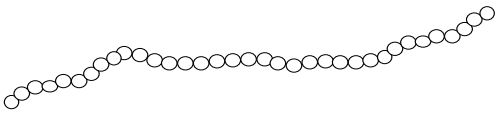
1. Improve hardness and stiffness.
2. Increase wear and solvent resistance.
3. Increase crazing resistance.
4. Increase brittleness.
5. Increase thermal resistance.
6. Decrease water sorption and solubility.

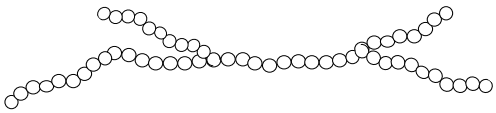


3. Monomer distribution in the polymer chain:

A. **<u>Homo polymers</u>** (one type of monomer) (linear or branched).

A-A-A-A-A-A

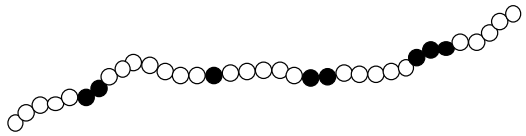




B. **<u>Co polymers</u>** (2 or 3 types of monomers) it is ether:

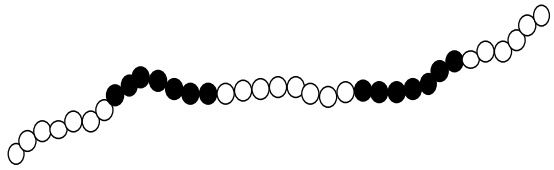
- *Random copolymer:* No sequential order exists among the two or more mer units along the polymer chain.

  …ABBABABAAABAAAABABBBBABAAAABABABB…

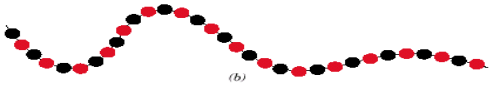


- *Block copolymer*: Identical monomer units occur in relatively long sequences (blocks) along the main polymer chain.

  …AAAAABBBBBBBAAAABBBBBBBAAABBBAAAA…

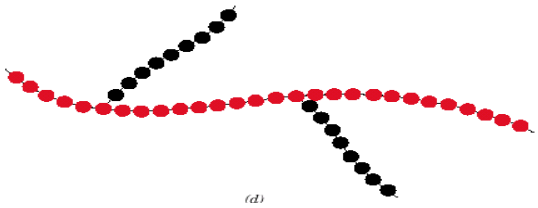


- *Alternating copolymer:* **…..ABABABABAB…..**



- *Graft or branched copolymer*: Sequences of one type of mer unit (B) are "grafted" onto a backbone chain of a second (A) type of mer unit to form a branched configuration.

… AAAAAAAAAAAAAA …
| |
B B
B B



4. Polymer molecular weight (Mw).

Mw of polymer molecules= the Mw of the mers x number of mers

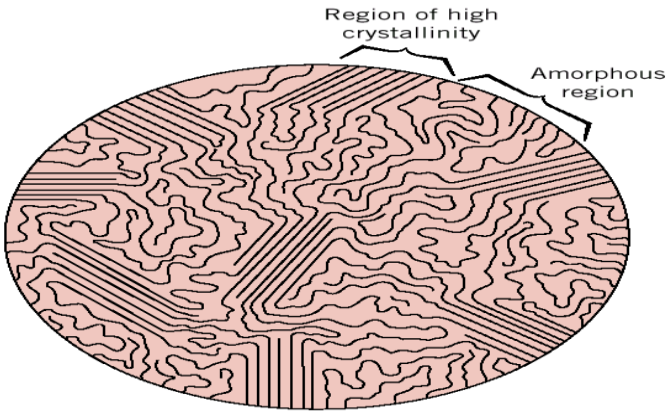
Mw may range from thousand to millions of molecular weight units depending on preparation conditions.

Mw of polymers plays an important role in determining its *physical properties*. Most polymers have a wide range of Mw and so vary widely in their properties. The higher the Mw the higher the softening and melting point and the stiffer the plastic. The higher the Mw of polymer made from a single monomer, the higher the degree of polymerization. The strength of the resin increases with the increasing of the degree of polymerization until a certain Mw is reached. Above this there is no change.

5. Supra molecular structure (molecular organization).

A. **Amorphous polymers** (coiled irregular) random Shape of polymer chains.

B. **Semi crystalline polymer:** domains with regular (crystalline) structure acting as special type of cross - links.



Polymers used in dentistry:

- **Natural polymers**. e.g. gutta percha.
- **Synthetic polymers** (prepared by polymerization reaction).

Application of polymers in dentistry:

1. **Prosthodontics:** denture bases, denture teeth, soft liners, custom trays, impression materials, cementing materials, maxillofacial prostheses …….
2. **Operative Dentistry:** dentin bonding agents, cavity fillings, resin and glass-ionomer cements, veneers…..
3. **Orthodontics:** brackets, bracket bonding resins and cements, and spacers.
4. **Endodontics:** gutta-percha points, root canal sealants, and rubber dams.
5. **Equipment:** mixing bowls, mouth guards , protective eyewear….

The types of polymerization reaction:

1. **Condensation polymerization.**
2. **Addition polymerization.**

1. **Condensation polymerization:** usually more than one type of monomer is used; The condensation reaction progress by the same mechanism as chemical reaction between two or more simple molecules. The reaction produces *by products* such as water, halogen acids and ammonia or alcohol.

2. **Addition polymerization:** *most dental resins* are polymerized by additional polymerization which simple involves the joining together of monomer molecules to form polymers chain. In this type of reaction, no by product is obtained. There is no change in composition during addition polymerization. It's formed from smaller units or monomers without change in composition, the monomer must have an *unsaturated group* in its structural formula for covalent bond with another molecule.

The additional reaction take place in 3 stages as follow:

1. Induction stage
2. Propagation stage
3. Termination stage

1. Induction stage: Two processes control the induction stage *(activation and initiation)*. For an addition polymerization process to begin, a source of *free radicals* (R•) is required. Free radicals can be generated by the activation of radical-producing molecules like **heat**, **visible light** and **chemical agents**.

Free radicals: are very reactive chemical species that have unpaired electron; the free radicals are produced by reactive agents called initiators. The free radical reacts with a monomer and initiates the polymerization process.

Initiators: are molecules which contain one relatively weak bond which is able to undergo decomposition to form two reactive species (free radicals). The decomposition of bond of initiator needs source of energy (activator) such as heat, light and chemical activator. Initiator which is used extensively in dental polymers is **Benzoyl peroxide**. Decomposed and free radicals are formed.

$$\underset{\text{Initiator}}{\mathbf{R}\text{—}/\text{—}\mathbf{R}} \xrightarrow{\textbf{+ external energy}} \underset{\text{free radicals}}{\mathbf{2\ R\bullet}}\ \text{(R= represent any organic molecular)}$$

$$\underset{\text{Free radicals}}{\mathbf{R\bullet}} + \underset{\text{monomer}}{\mathbf{M}} \longrightarrow \underset{\text{reactive free radical species}}{\mathbf{R\text{-}M\bullet}}$$

The initiation reaction is an addition reaction producing another active radical species which is capable of further reaction (addition of free radical on the double bond of monomer).

The initiation period depends on the purity of the monomer and the temperature of the reaction. Any impurities present that are able to react with activated groups can increase the length of this period (induction period). The higher the temperature, the more rapid the formation of free radicals and consequently the shorter the induction period.

2. Propagation stage: The initiation stage is followed by the rapid addition of other monomer molecules to the free radical and shifting of free electron to the end of growing chain, this called propagation stage.

R—M• + M ⟶ R—M—M•
R—M—M• + M ⟶ R—M—M—M•
R—M—M—M• + M ⟶ R—M—M—M—M•

3. Termination stage: The growing chain is stopped. Termination occurs when monomer units are used up or free radical is finished by reaction. Chain reaction can be terminated either by reaction of two growing chains to form one dead chain or by reaction of growing chains with molecules of initiation, dead polymer, impurity or solvent if present.

R—M—M—M• + R—M—M—M• ⟶ R—M—M—M—M —M—M— R

Inhibitor: it is chemical materials added to prevent or delay polymerization during storage and in order to provide enough working time and decrease sensitivity to ambient light like **hydroquinone**.

The following factors inhibit the polymerization:

1. Any **impurity** in the monomer that can react with free radicals or with any activated growing chain to prevent further growing.
2. The addition of small amounts of **hydroquinone** to the monomer.
3. The presence of **oxygen** also causes retardation of polymerization reaction because the oxygen reacts with free radicals and so the degree of polymerization is decreased.

Plasticizer: these are substance added to the resins to:

1. Decrease the brittleness of the polymer.
2. Increase the solubility of the polymer in the monomer.
3. It decreases strength, hardness and softening point it is used to prepare flexible polymer.

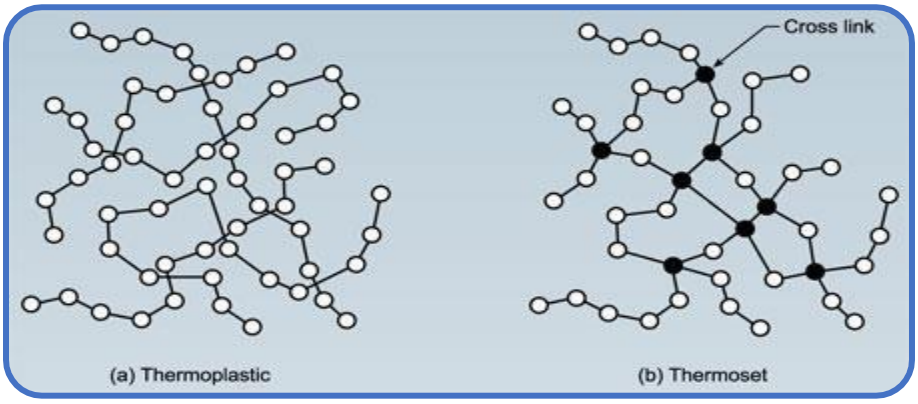
Classification of polymer according to thermal response:

1. Thermoset polymer.
2. Thermoplastic polymer.

Thermosetting polymer or resins: Thermosetting polymers undergo a chemical change during the setting reaction and become permanently hard when heated above the temperature at which they begin to polymerize; they do not soften again on reheating to the same temperature. They are usually cross linked in this state, and thus, they are insoluble and will not melt. Instead, they decompose if heated to a high enough temperature. Thermosetting polymers generally have superior abrasion resistance and dimensional stability compared with thermoplastic polymers.

Thermoplastic polymer or resins: They soften when heated above the glass transition (The resin can be shaped and molded) and upon cooling it will harden; this cycle can be repeated almost indefinitely. The setting

reaction is reversible because of the relatively weak bonds among the molecular chains. Thus, in contrast to thermosetting resins, thermoplastic resins can be melting and are usually soluble in organic solvents and have better flexural and impact properties. e.g. flexible denture.



Denture base resin material

Various materials have been used to construct dentures like cellulose products, phenol- formaldehyde, vinyl resins and vulcanite, however they have suffering from a variety of problems. Acrylic resin (polymethyl methacrylate) (PMMA) is now the material of choice to use as denture base material because it is easy to process and use, cheep and good esthetic. Even so it is not ideal in all respects.

The ideal requirements of denture base material represents by the followings:

1. Natural appearance.
2. High strength, stiffness, hardness and toughness.
3. Dimensionally stability.
4. Tasteless, odorless, non toxic and none irritating the oral tissue.
5. Insoluble in saliva or other fluids taken in to the mouth.
6. Good retention to polymer, porcelain and metals.
7. Easy to fabricate and Easy to repair.
8. Good shelf life.
9. Low density.
10. Accurate reproduction of surface details.
11. Resistance to bacterial growth.
12. Good thermal conductivity.
13. radiopacity
14. Easy to clean
15. Inexpensive to use
16. The softening temperature should be higher than mouth temperature.

The Types Of Denture Base Resin:

- **Heat cure** denture base resin, set by heat
  a. water bath
  b. microwave
- **Light cure** denture base resin
- **cold cure, self-cure** denture base resin

Classification of denture base materials

According to the Type of Materials

- metallic e.g. Cr/Co denture base
- nonmetallic materials e.g. poly (methyl methacrylate)

According to the Method of Polymerization

- Addition polymers, e.g. poly (methyl methacrylate)
- Condensation polymers, e.g. nylon.

According to their Thermal Response

- **Thermoplastic**, e.g. polyvinyl acrylics and polystyrene.
- **Thermosetting**, e.g. poly (methyl methacrylate).

According to the Method of Activation

- ***Thermal***; e.g. poly (methyl methacrylate).
- ***Microwave***; e.g. poly (methyl methacrylate).
- ***Catalyst***; e.g. poly (methyl methacrylate).
- ***Visible light activated***; e.g. poly (methyl methacrylate).

According to the Dispensing System

- **Power and liquid system**, e.g. heat-cure and self-cure acrylic resins.
- **Sheets and ropes**: (Single component system), e.g. light activated denture base resins.
- **Gel type**, e.g. vinyl acrylics.